

MSE 3114 Fall 2025 Lesson 1: Setup and First Basic Data Analysis

Overview

Duration: 2 weeks (Weeks 1-2)

Weekly Workload: 3-4 hours

Learning Focus: Anaconda environment setup, AI tool configuration and first stress-strain analysis

Learning Objectives

By the end of this lesson, you will be able to:

- **Install Anaconda Python and required packages**
 - **Configure Anaconda environment**
 - **Configure AI tools** for materials science research
 - **Load and validate** stress-strain data using Python
 - **Perform basic analysis** with AI assistance
 - **Create simple visualizations** of mechanical properties
 - **Understand AI tool limitations** and best practices
-

SCREEN CLIP INSTRUCTIONS

Throughout this lesson, you'll see this icon: 

When you see , you need to take a screenshot or screen recording to document your progress.

How to take screen clips:

- **Windows:** Press `Windows key + Shift + S` for snipping tool
- **macOS:** Press `Cmd + Shift + 4` for area selection
- **Linux:** Press `PrtScn` or use built-in screenshot tools

Save your clips in a pdf file with descriptive captions like:

- `Cursor testing`
- `github connection`

- python version
 - etc
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Week 1: AI Tools Setup and Anaconda Python Environment Configuration

Introduction to AI-Augmented MSE 3114

Welcome to MSE 3114! This course will teach you how to use artificial intelligence tools to enhance your materials science research. We'll start by setting up the essential AI tools and then apply them to a classic materials science problem: stress-strain analysis.

Why AI-Augmented Materials Science?

Traditional materials science research involves:

- Manual data analysis and interpretation
- Time-consuming literature reviews
- Limited experimental design optimization
- Basic statistical analysis

AI augmentation provides:

- **Accelerated Analysis:** Faster data processing and interpretation
- **Enhanced Insights:** Pattern recognition beyond human capability
- **Optimized Design:** AI-assisted experimental planning
- **Automated Reporting:** Quick generation of analysis summaries

Essential AI Tools Setup

Installation Order: The order matters! Install tools in this sequence:

- I. **Anaconda** (Python environment)
- II. **GitHub Account** (for Copilot access)
- III. **Cursor** (AI development environment)

I. Anaconda Python Environment Setup

Anaconda Update (if you already have Anaconda installed)

Step 1: Open Anaconda Prompt

- Click the **Start** button (Windows icon)
- Type "Anaconda Prompt"
- Click on "**Anaconda Prompt (anaconda3)**" or "**Anaconda PowerShell**"

Step 2: Update Conda (the package manager) In the Anaconda Prompt window, type this command and press **Enter**:

```
conda update conda
```

- When prompted, type **y** and press **Enter** to confirm
- Wait for the update to complete

Step 3: Update All Packages After Step 2 completes, type this command and press **Enter**:

```
conda update --all
```

- When prompted, type **y** and press **Enter** to confirm
- This may take several minutes - be patient!

Step 4: Clean Up (Optional but Recommended) After all updates complete, type this command and press **Enter**:

```
conda clean --all
```

- When prompted, type **y** and press **Enter** to confirm

Step 5: Verify the Update Type this command to check your conda version:

```
conda --version
```

Step 6: Close the Window

- Type **exit** and press **Enter**
- Or simply close the Anaconda Prompt window

What This Does

- **Step 2:** Updates the conda package manager itself
- **Step 3:** Updates all your installed packages to their latest compatible versions
- **Step 4:** Removes old package files to free up disk space

If Something Goes Wrong

- Don't panic - your old packages are still there
- Close the Anaconda Prompt and try again
- If problems persist, ask your instructor for help

Anaconda Installation (For clean installs.)

Step 1: Download Anaconda

1. Visit <https://www.anaconda.com/>
2. Click "Download" button
3. Choose your operating system:

- **Windows:** Download the 64-bit installer (.exe file)
- **macOS:** Download the 64-bit graphical installer (.pkg file)
- **Linux:** Download the 64-bit installer (.sh file)

Step 2: Install Anaconda

- **Windows:** Double-click the .exe file and follow the installation wizard
 - **Important:** Check "Add Anaconda to PATH" during installation
 - **Important:** Check "Register Anaconda as default Python"
- **macOS:** Double-click the .pkg file and follow the installation wizard
- **Linux:** Open terminal and run: `bash Anaconda3-*.sh`

Step 3: Verify Anaconda Installation

Opening Command Prompt in Windows:

1. **Method 1 (Easiest):** Press `Windows key + R`, type `cmd`, press Enter
2. **Method 2:** Press `Windows key`, type `cmd`, click "Command Prompt"
3. **Method 3:** Right-click Start button, select "Windows Terminal" or "Command Prompt"

What You'll See:

- A black window with white text will open
- You'll see something like: `C:\Users\YourName>`
- This is called the "command prompt" or "terminal"

Verify Anaconda Installation:

1. In the command prompt, type exactly: `conda --version`
2. Press Enter
3. You should see something like: `conda 23.x.x` or `conda 24.x.x`
4. 📷 *Take screenshot of your output*

If You See "conda is not recognized":

- Close the command prompt
- Restart your computer (this ensures PATH is updated)
- Try again with a new command prompt

If You Still Get Errors:

- Make sure Anaconda was installed with "Add to PATH" checked
- Try installing Anaconda again with the correct options

In []:

Install Required Packages (For both the previous installations.)

Important: Use Anaconda Prompt, not regular Command Prompt!

Opening Anaconda Prompt:

1. **Method 1:** Press `Windows key` , type `Anaconda Prompt` , click "Anaconda Prompt (anaconda3)"
2. **Method 2:** Press `Windows key` , type `Anaconda` , click "Anaconda Navigator" → "Environments" → "Open Terminal"
3. **Method 3:** Start Menu → Anaconda3 → Anaconda Prompt

What You'll See:

- A blue/black window with `(base)` at the beginning
- Example: `(base) C:\Users\YourName>`
- This means you're in the base Anaconda environment

At Anaconda Prompt type:

```
# Install core scientific packages using conda
conda install pandas numpy matplotlib scipy scikit-learn
```

```
# When prompted "Proceed ([y]/n)?", type 'y' and press Enter
# Wait for installation to complete
```

```
# Install additional packages using pip
pip install opencv-python plotly streamlit polars duckdb
```

```
# Wait for all packages to install
```

Step 5: Verify Package Installation

```
# Check that all packages are installed
conda list
# and check for pandas numpy matplotlib scipy scikit-learn in the list
pip list | findstr "opencv plotly streamlit polars duckdb"
```

Test Installation

Use Jupyter Lab (Recommended for beginners)

Opening Jupyter Lab:

1. Open your Anaconda PowerShell terminal
2. Navigate to your project directory: `cd C:\path\to\your\project`
3. Launch Jupyter Lab: `jupyter lab`
4. Your browser will open automatically with Jupyter Lab

What You'll See:

- A web page opens with Jupyter Lab interface

- Click the "+" button to create a new notebook
- Choose "Python 3" when prompted
- You'll see a new notebook with code cells

Test Your Environment: Copy and paste this code into a Jupyter notebook cell:

```
# Test your environment - copy and paste this code
import pandas as pd
import numpy as np
import matplotlib
import matplotlib.pyplot as plt
import cv2
import plotly
import streamlit as st
import polars as pl
import duckdb

print("=== PACKAGE IMPORT TEST ===")
print("All packages imported successfully!")
print(f"Pandas version: {pd.__version__}")
print(f"NumPy version: {np.__version__}")
print(f"Matplotlib version: {matplotlib.__version__}")
print(f"OpenCV version: {cv2.__version__}")
print(f"Plotly version: {plotly.__version__}")
print(f"Streamlit version: {st.__version__}")
print(f"Polars version: {pl.__version__}")
print(f"DuckDB version: {duckdb.__version__}")

print("\n=== BASIC FUNCTIONALITY TEST ===")
# Test basic functionality
data = pd.DataFrame({'x': [1, 2, 3], 'y': [4, 5, 6]})
print("Pandas DataFrame created successfully:")
print(data)

# Test plotting
plt.figure(figsize=(6, 4))
plt.plot(data['x'], data['y'], 'o-')
plt.title('Test Plot')
plt.xlabel('X values')
plt.ylabel('Y values')
plt.show()

print("Environment setup complete! 🎉")
```

Running Code in Jupyter Lab:

- Click in a code cell
- Press **Shift + Enter** to run the cell
- Or click the "Run" button (play icon)
- Run cells one at a time to see output

Expected Output: You should see:

- All packages import without errors
- Version numbers displayed for each package
- A simple test plot displayed
- "Environment setup complete! 🎉" message
- 📸 *Take screenshot of your output*

If You Get Errors:

- Check that all packages were installed successfully
- Try restarting your terminal/command prompt

II GitHub Account Setup (Optional for Copilot)

Step 1: Create GitHub Account

1. Visit github.com
2. Click "Sign up" and create a free account
3. Verify your email address
4. **Important:** Apply for [GitHub Student Developer Pack](#) if you're a student
 - This gives you free access to GitHub Copilot and other developer tools
 - Use your VT email address for verification

III. Cursor (Primary AI Development Environment) Setup

Purpose: All-in-one AI development environment with built-in AI features

Why Cursor? Cursor is built on VS Code but includes AI features built-in, making it ideal for this course.

Setup Steps:

1. **Download Cursor:** Visit cursor.sh
2. **Install Cursor:** Run the installer for your operating system
3. **Sign in with GitHub:** This gives you access to GitHub Copilot automatically
4. **Install Github Extension:**
 - Go to Extensions (Ctrl+Shift+X)
 - Search for "GitHub Pull Requests" by GitHub
 - Click Install
5. **Restart Cursor and log back in with GitHub**
6. **Check that GitHub is working**
 - In the terminal window in Cursor type: `git --version`. You should see git with its version displayed.
7. **Install Python Extension:**
 - Go to Extensions (Ctrl+Shift+X)
 - Search for "Python" by Microsoft

- Click Install

8. **Verify AI Features:**

- Press Ctrl+K to open AI chat
- Press Ctrl+L to open AI command palette

Cursor Advantages for This Course:

- Built-in AI chat and code generation
- Seamless GitHub Copilot integration
- AI-powered code explanation and debugging
- Better AI context understanding
- Free for students (GitHub Student Developer Pack)

B. Connect to Python (Required for AI Code Generation):

1. Install Python Extension (if not already done):

- Press `Ctrl+Shift+X` to open Extensions
- Search for "Python" by Microsoft
- Click Install if not already installed

2. Select Python Interpreter:

- Press `Ctrl+Shift+P`
- Type "Python: Select Interpreter"
- Choose "Python 3.x.x ('base')"

3. Install Jupyter Extension

- Press `Ctrl+Shift+X`
- Type "Jupyter" by microsoft
- Click Install

4. Close and restart Cursor

5. Verify Python Connection:

- **Open or Create a python notebook** in Cursor
 - `Ctrl+Shift+P` type "Create:New Jupyter Notebook" then select.
- **Select Interpreter** Press `Ctrl+Shift+P` , type "Python: Select Interpreter". Pick "Python 3.x.x ('base')"
- **Check top-right corner:** With your notebook open in Cursor look in the top-right corner. It might say "Kernel" or "Select Kernel". If so click on it, "python 3.x.x ('base')" if available. If it isn't available click on "python environment" and then you should see it.
- 📸 **Take screenshot showing Python interpreter selection in top-right corner**

C. Test AI Code Generation:

- In the center panel (your new file), type: `# Create a pandas DataFrame with sample data` and press enter to move to the next line.
- Press `Ctrl+K` to open AI chat
- Ask: "Help me create a sample DataFrame with 5 rows of random data"
- The AI should generate code using your installed packages.

First AI Interaction: Stress-Strain Analysis Planning

Let's use AI to help plan our first analysis. This demonstrates how AI can assist in research planning. In the AI chat panel on the right (in Cursor), paste the template given below.

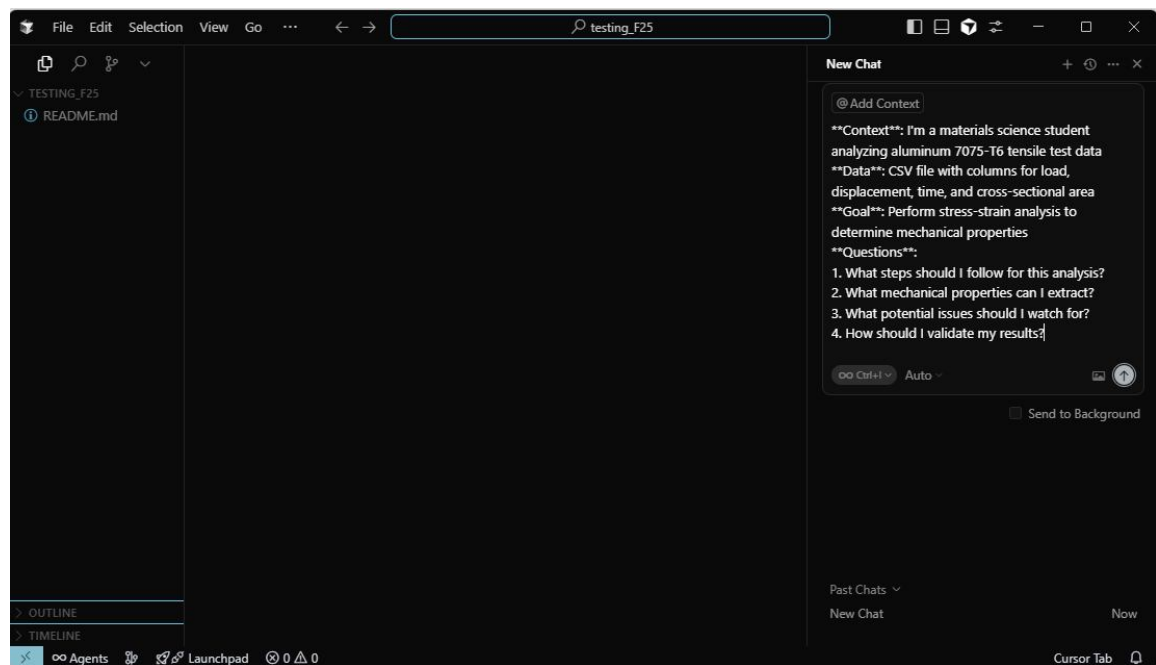
AI Prompt Template

```

**Context**: I'm a materials science student analyzing aluminum
7075-T6 tensile test data
**Data**: CSV file with columns for load, displacement, time, and
cross-sectional area
**Goal**: Perform stress-strain analysis to determine mechanical
properties
**Questions**:
1. What steps should I follow for this analysis?
2. What mechanical properties can I extract?
3. What potential issues should I watch for?
4. How should I validate my results?

```

Here is my screenshot of my Cursor interface. Notice I just pasted the above template straight into Cursor. The formatting codes such as the `"**"` etc won't hurt anything.



Expected AI Response

The AI should provide:

- Step-by-step analysis workflow
- List of mechanical properties to calculate
- Common data quality issues to check
- Validation methods for results

Week 1 Assignment: Environment Setup

Due: End of Week 1

Points: 10 points

Deliverables:

1. **All snippets images** from the lesson above with single sentence descriptions.
2. **AI interaction log** with your first stress-strain analysis prompt
3. **Comment** What surprised you about the AI's response?
4. **Create** a new prompt to give you "better" results?
5. **Comment** How did your improved prompt change the output?
6. **Comment** What would you do differently next time?
7. **AI interaction log** of your new edited stress-strain analysis prompt.

Submission Format: pdf file upload in Canvas

In []: